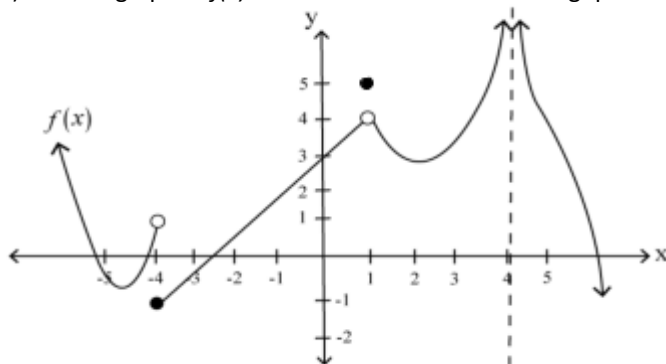


Show work if applicable. Use infinite limits to describe unbounded behavior.

1. (2pts) Use the results presented in the table to evaluate the following limit: $\lim_{x \rightarrow -3} \frac{\sqrt{1-x}-2}{x+3} =$ _____

x	- 3.1	- 3.01	- 3	- 2.99	- 2.9
$f(x)$	- 0.2485	- 0.2498	??	- 0.2502	- 0.2516

2. (8pts) Use the graph of $f(x)$ below to answer the following questions.



a. $\lim_{x \rightarrow -4^-} f(x) =$

b. $\lim_{x \rightarrow -4^+} f(x) =$

c. $\lim_{x \rightarrow 4} f(x) =$

d. $\lim_{x \rightarrow 1} f(x) =$

e. $\lim_{x \rightarrow 4} f(x) =$

f. $\lim_{x \rightarrow \infty} f(x) =$

- g. Is $f(x)$ continuous at $x = 4$? Use the definition of continuity to explain why or why not.

3. (3pts) Use $\lim_{x \rightarrow c} f(x) = 2$ and $\lim_{x \rightarrow c} g(x) = -3$ to find the following:

a. $\lim_{x \rightarrow c} [f(x) - 3g(x)] =$

b. $\lim_{x \rightarrow c} [f(x) \cdot g(x)] =$

c. $\lim_{x \rightarrow c} \left[\frac{f(x)}{g(x)} \right] =$

4. (4pts) Find the limit algebraically (if it exists) as x approaches ∞ .

a. $\lim_{x \rightarrow \infty} \sin\left(\frac{\pi x}{2}\right)$

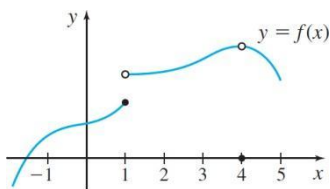
b. $\lim_{x \rightarrow \infty} \frac{1}{4} - \frac{1}{4+x}$

5. (3pts) Find a so that the function is continuous for all real values of x .

$$f(x) = \begin{cases} 2x - a & \text{if } x < -3 \\ ax - 6 & -3 \leq x \end{cases}$$

6. (4pts) Verify that the Intermediate Value Theorem applies. If it does, use it to show that $f(x) = x^2 - 4x + 3$ has a zero on the interval $[2, 4]$. Then find the value(s) of c guaranteed by the theorem.

7. (2pts) The graph of a piecewise function f is shown.



Use the graph to determine which of the following statements is true.

- (A) $\lim_{x \rightarrow -1^+} f(x)$ does not exist
- (B) $\lim_{x \rightarrow 1^-} f(x) = f(1)$
- (C) $\lim_{x \rightarrow 1} f(x)$ exists
- (D) $\lim_{x \rightarrow 4} f(x) = f(4)$
8. (2pts) For $x > 0$, the line $y = 1$ is an asymptote of the graph of a function f . Which of the following statements must be true?
- (A) $f(x) \neq 1$ for $x > 0$
- (B) $\lim_{x \rightarrow 1} f(x) = \infty$
- (C) $\lim_{x \rightarrow \infty} f(x) = 1$
- (D) $\lim_{x \rightarrow -\infty} f(x) = 1$